

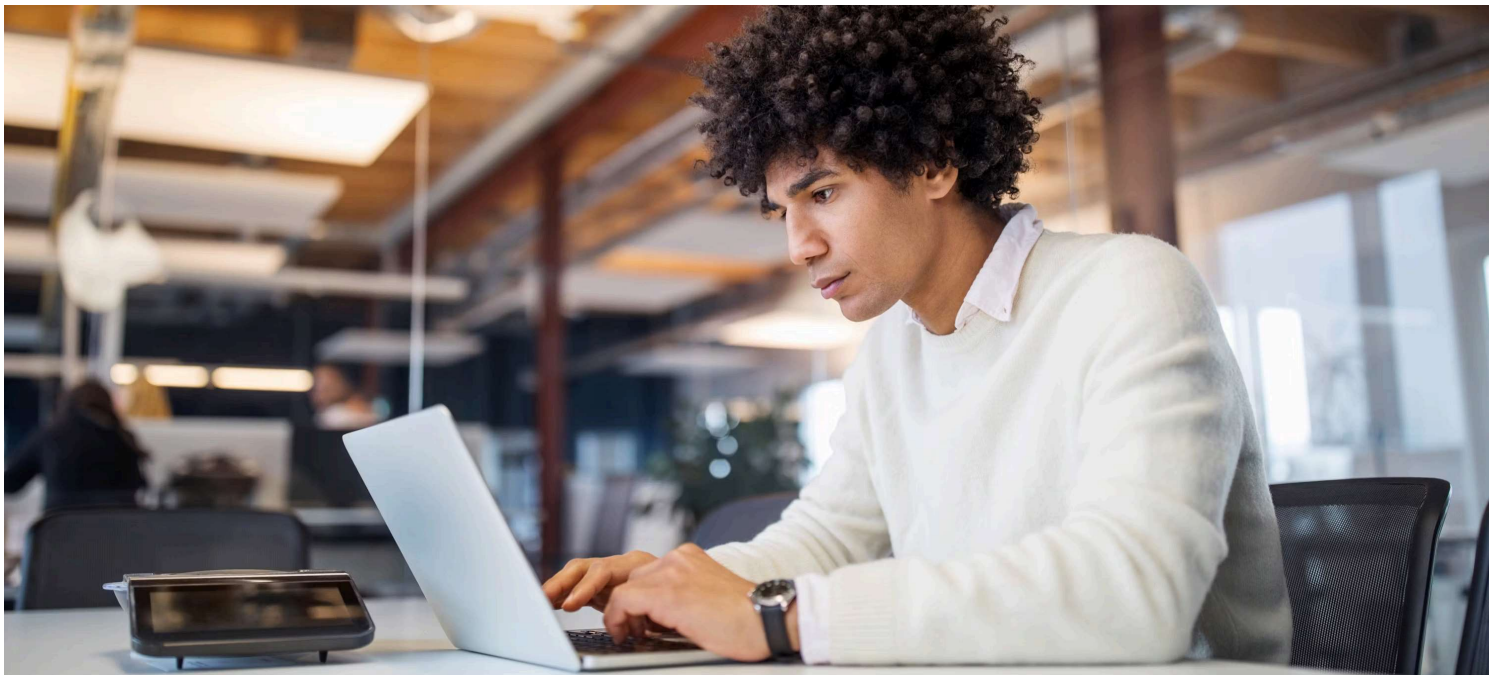
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What Is Generative AI? Definition, Applications, and Impact

Written by Coursera Staff • Updated on Apr 3, 2024

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Generative artificial intelligence (GenAI) can create certain types of images, text, videos, and other media in response to prompts. Here's what you should know about this growing field and tool.



Generative artificial intelligence (AI) is a type of AI that generates images, text, videos, and other media in response to inputted prompts.

AI generators like ChatGPT and DALL-E2 are gaining worldwide popularity. These programs respond to prompts input by users. Submit a text prompt, and the generator will produce an output, whether it is a story or outline from ChatGPT or a monkey painted in a Victorian style by DALL-E2.

In this article, we'll outline what you should know about this growing field, how it works, uses cases, and more. Afterward, if you're interested in building foundational generative AI skills, consider enrolling in DeepLearning.AI's Generative AI for Everyone course.



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What is generative AI?

Generative AI, also referred to as GenAI, allows users to input a variety of prompts to generate new content, such as text, images, videos, sounds, code, 3D designs, and other media. It “learns” and is trained on documents and artifacts that already exist online.

Generative AI evolves as it continues to train on more data. It operates on AI models and algorithms that are trained on large unlabeled data sets, which require complex math and lots of computing power to create. These data sets train the AI to predict outcomes in the same ways humans might act or create on their own.

Read more: [Artificial Intelligence \(AI\) Terms: A to Z Glossary](#).

The rise of generative AI is largely due to the fact that people can use natural language to prompt AI now, so the use cases for it have multiplied. Across different industries, AI generators are now being used as a companion for writing, research, coding, designing, and more.



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How does generative AI work?

Generative AI models use neural networks to identify patterns in existing data to generate new content. Trained on unsupervised and semi-supervised learning approaches, organizations can create foundation models from large, unlabeled data sets, essentially forming a base for AI systems to perform tasks [1].

Some examples of foundation models include LLMs, GANs, VAEs, and Multimodal, which power tools like ChatGPT, DALL-E, and more. ChatGPT draws data from GPT-3 and enables users to generate a story based on a prompt. Another foundation model Stable Diffusion enables users to generate realistic images based on text input [2].

Hear more about how generative AI works in this lecture from DeepLearning.AI's [Generative AI for Everyone](#) course:

Andrew Ng explains how generative AI works.



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Popular AI generators

There are several generative AI platforms you can become familiar with. You may find them helpful for automating certain processes in your workflow.

- **ChatGPT:** This language model has a foundation of GPT architecture that generates text that resembles something a human would produce. It's a helpful companion for research, strategy, and content creation.

- **DALL-E2:** This model generates images from text prompts, so creatives can create vibrant illustrations and concept art that's a useful accompaniment to content marketing.
- **GitHub Copilot:** This collaboration between GitHub and OpenAI acts as a coding companion to help developers code faster and more intuitively.

Read more: [6 Artificial Intelligence \(AI\) Jobs to Consider in 2023](#)



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Generative AI use cases

Once you've decided which AI generator suits your needs, these use cases may help you get the creative juices flowing for ways generative AI can benefit you and your business.

- Writing or improving content by producing a draft text in a specific style or length
- Adding subtitles or dubbing educational content, films, and other content in different languages
- Outlining briefs, resumes, term papers, and more
- Receiving a generic code to edit or improve upon
- Summarizing articles, emails, and reports
- Improving demonstration or explanation videos
- Creating music in a specific tone or style

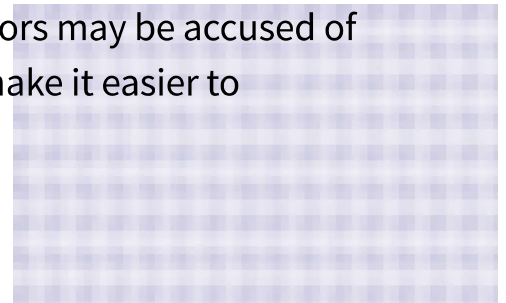
Generative AI has many use cases that can benefit the way we work, by speeding up the content creation process or reducing the effort put into crafting an initial outline for a survey or email. However, generative AI also has limitations that may cause concern if it remains unregulated.



Concerns about generative AI

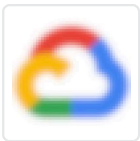
Generative AI's popularity is accompanied by concerns of ethics, misuse, and quality control. Because it is trained on existing sources, including those that are unverified on the internet, generative AI can provide misleading, inaccurate, and fake information. Even when a source is provided, that source might have incorrect information or may be falsely linked.

Since generators such as ChatGPT allow humans to input prompts with everyday language, it has become easier to use--so much so, that university students might use it to plagiarize or generate essays, and content creators may be accused of stealing from original artists. Falsified information can make it easier to impersonate people for cyber attacks.



Get an intro to generative AI

For a quick, one-hour introduction to generative AI, consider enrolling in Google Cloud's [Introduction to Generative AI](#). Learn what it is, how it's used, and why it is different from other machine learning methods.



Introduction to Generative AI Course

Beginner level · 1 hour(s)

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To get deeper into generative AI, you can take IBM's [Generative AI Fundamentals Specialization](#) course and learn the potential of this new AI technology. This course is designed for beginners with no prior experience and covers topics like prompt engineering, generative AI careers, foundation models, and more.

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